Valhall Re-Development Project



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T33-02	23	MB	PH-11530-O-4090008	1

Installation, Operation and Maintenance Manual for

1st Stage Discharge Cooler 23-HA-9114/9115

according to the European Pressure Equipment Directive (PED) Directive 97/23/EG

Content Page

1. General 3
2. Further Directives of the European Community 3
   2.1 Directive 94/9/EC (ATEX) 3
   2.2 Further Directives of the European Community 3
3. Design data for 23-HA-9114/9115 4
4. Transportation and arrival at site 5
5. Installation and preparation for putting in service 6
6. Start up, operation and shut down 7
7. Maintenance and Inspection 8

1. General

Installation, Operation and Maintenance Manual is prepared to supply the user of the 1st Stage Discharge Coolers 23-HA-9114 and 23-HA-9115 with the necessary information according to the Pressure Equipment Directive (PED) concerning the operation and the residual hazards resulting from the pressure. These operating instructions do not replace the instructions issued by the operator of the whole facility where the Heat Exchanger is installed.

Important Note:
This Heat Exchanger is allowed to be installed, put in service and operated by trained, competent and experienced personal only. The training, the competence and the experience shall be related to the installation, the putting in service and the operation of Heat Exchangers and the possible hazards that go out from Heat Exchanger especially from the pressurisation. Appropriate transportation and storage are preconditioned.

The following documents are required for the use and to understand these Installation, Operation and Maintenance Manual. The documents shall be available when reading these instructions and shall be taken into account. The General Arrangement drawing describes the Heat Exchanger in detail.

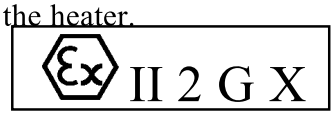
Documents referred to in the operation instructions for Tag No.: 23-HA-9114/9115

Document name	Leffer Doc. No.	Project document no.
Gen. Arr.- Dwg.	300908 G 01	PH-11530-R-4090019-001
Shell Detail Dwg	300908 S 01	PH-11530-R-4090020-001
Bundle Detail Dwg 1	300908 B 01	PH-11530-R-4090021-001
Nozzle Detail Dwg	300908 N 01	PH-11530-R-4090022-001
Saddle Detail Dwg	300908 S 02	PH-11530-R-4090036-001
Shipping Dwg	300908 T 01	PH-11530-O-4090007-001
Spare Parts List and Interchangeability Record - SPIR	300906 SL 01	PH-11530-O-4090001

2. Further Directives of the European Community

2.1 Directive 94/9/EC (ATEX)

The purchaser advises that the heat exchanger is located in Zone 1 according to Directive 94/9/EC (ATEX). The pressure inside the heater is above the atmospherical pressure, therefore the ATEX directive is not applicable inside the heater. 1

The cooler is marked:  to inform the user about its suitability to be used in zone 1 according to the ATEX directive 1

Indication of Information which is essential for safe use of this equipment
(advised by the letter X in the above marking):
The surface temperature of the coolers is determined by the media inside (Natural Gas on the Tube Side and Cooling Medium on the Shell Side). The maximum surface temperature is +175,0°C according to the data given in Chapter 3 of this document.
This document covers in the following chapters all aspects to avoid that the heaters will become an ignition source or to avoid that the ATEX directive is applicable for the inside (maximum allowed temperature, earthing, venting/inertisation, etc.).

2.2 Further Directives of the European Community

Further directives of the European Community are not applicable for this pressure equipment.

3. Design data for 23-HA-9114/9115

Manufacturer	**Hans Leffer GmbH & Co. KG**
Conformity assessment module	**Module H1**
Design rule	**PD 5500:2006 Cat.1**

Shell side:

Maximum design pressure	**15,8 bar g**	Minimum design pressure	**F.V.**
Test pressure	**27,3 bar g**		
Maximum design temperature	**+150°C**	Minimum design temperature	**-8,2°C**
Volume	**5900 l**		

Tube side:

Maximum design pressure	**30,0 bar g**	Minimum design pressure	**F.V.**
Test pressure	**53,8 bar g**		
Maximum design temperature	**+175°C**	Minimum design temperature	**-28°C**
Volume	**6400 l**		

For Outline Dimensions see GA drawing (PH-11530-R-4090019-001)

Further loadings according to Project Specifications

Load	Taken into account	Not taken into account
external loads		
1. Nozzle Loads acc. to specification	X	
2. Blast load		X
3. Wind load acc. to specification	X	
4. Transportation loads acc. to specification	X	
5. Further external loads		X
Internal loads (1	X	
Earthquake		X
Vibrations (2	X	
External fire		X

Notes: 1) These loads are resulting from the operation of the heat exchanger

2) Tube Vibrations are taken into account by a design which avoids their occurrence during operation.

4. Transportation and arrival at site

Check

The Heat Exchanger was thoroughly checked to be complete and undamaged before leaving the works of Hans Leffer GmbH & Co. KG. It was further checked that all accessories and spare parts according to the order and the shipping drawing are shipped with the Heat Exchanger.

see:
- GA-drawing
- Shipping-drawing

Installation / Mounting

The Heat Exchanger will be equipped with two lifting lugs on the top of the shell. They allow for lifting of the complete Heat Exchanger. Lifting of the Heat Exchanger is permitted only by using certified slings and shackles and attaching those to the lifting lugs.
The lifting lugs on the channels are for lifting of the channels only.
The Heat Exchanger is provided with two earthing lugs. To dissipate static electricity the Heat Exchanger shall be connected to ground at these two earthing lugs by suitable equipment.

see:
- GA-drawing
- Shipping-drawing

Protection of gasket surfaces during transportation and storage

All nozzles that are ordered with a blind flange are supplied with original bolts and gaskets and original blind flange. The other nozzle flanges are closed by a temporary steel cover, temporary bolts and temporary rubber gaskets.

see:
- GA-drawing
- Nozzle Detail drawing
- Shipping-drawing

Check for completeness

After arrival at site the user must check that the Heat Exchanger is complete and undamaged. The User must further check that all accessories and spare parts are attached according to the purchase order and the packing list.

see:
- Packing list
- Spare parts list

Check of damages

After arrival at site the Heat Exchanger shall be visually checked for any damages on the surface. If such damages are found Hans Leffer GmbH & Co. KG shall be informed. The damaged Heat Exchanger shall not be put in operation without written approval of Hans Leffer GmbH & Co. KG.

visual check

5. Installation and preparation for putting in service

Mounting and fixing

The user is responsible for the correct installation of the Heat Exchanger on a suitable dimensioned foundation.
The Heat Exchanger will be operated in a horizontal position. It is provided with two saddles.
The Heat Exchangers will be mounted stacked each over. The saddles of the bottom exchanger (item 23-HA-9114) shall be welded to the foundation. Top exchanger (item 23-HA-9115) shall be installed on bottom exchanger. Only the bolts according to the drawing shall be used. A torque moment for these bolts is not required. “Hand tightening” is sufficient. Shim plates according to drawing shall be used.

see:
- GA-drawing

The Heat Exchanger must be horizontally aligned on its foundation. If required shim plates shall be used. The deviation of height for the two foundations shall be below 2 mm.

Nozzle / Piping connections

To avoid pollution and moisture getting into the Heat Exchanger as well as damaging of the gasket surfaces all open nozzles are provided with temporary steel covers, temporary bolts and an aluminised tape. These protection devices shall not be removed until the piping will be connected to the Heat Exchanger.
Only the gaskets supplied with the coolers or gaskets ordered acc. to the Spare Parts List are permitted. The user has to make sure that the gaskets are suitable for the medium intended to be inside the Heat Exchanger.

see:
- GA-drawing
- Shipping drawing
- Spare Parts List

Insulation Work

The heat exchanger is not insulated.
Adequate measures should be taken in order to avoid any accidental contact with the hot surfaces of the exchanger.

Repair by welding / flame cutting

It is not permitted to weld or flame cut on pressure carrying parts of the Heat Exchanger.

see:
- GA-drawing

6. Start up, operation and shut down

Start up

During start up excessive heating shocks of the material shall be avoided. The heater shall be filled with the cold medium at any stage of operation. In no case the exchanger shall be exposed to abrupt temperature changes (cold medium in hot exchanger/hot medium in cold exchanger).

Excessive local heating shall be avoided. Care shall be taken to get a uniform increase in temperature. In any stage of start up the Heat Exchanger shall not be exposed to abrupt changes of temperature and/or pressure.

Normal Operation

The Heat Exchanger is designed according to the design data in paragraph 3 which are given by the user/purchaser.

see
- GA-drawing

The maximum and minimum design temperatures are given on the Heat Exchanger name plate.
Safety accessories such as pressure limiting or temperature limiting devices are not part of the supply by Hans Leffer GmbH & Co. KG. The user must assure that the Heat Exchanger is operated in each state of operation inside the limits given by the maximum and minimum temperature and the maximum and minimum pressure.
The scope of supply of Hans Leffer GmbH & Co. KG does not include any insulation of the Heat Exchanger. Changes of pressure and temperature inside the Heat Exchanger shall not be abrupt.

Shut down

During shut down of the Heat Exchanger care shall be taken to get a uniform decrease in temperature. The heater shall be filled with the cold medium at any stage of operation.

Excessive local cooling shocks shall be avoided. In any stage of shut down the Heat Exchanger shall not be exposed to abrupt changes of temperature and/or pressure.

7. Maintenance and Inspection

Opening of the Channels

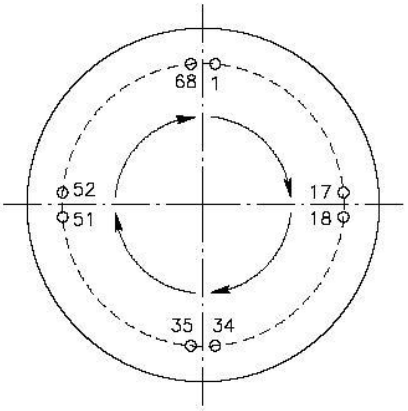
see:
- GA-drawing
- Shell detail drawing
- Spare parts list

Before opening the Channel the user must ensure that there is no differential pressure between the inside of the Channel and the outside of the Heat Exchanger. The Heat Exchanger must have cooled down to ambient temperature.
The user must further ensure that there are no hazardous fluids or gases can escape from the Heat Exchanger.
Before opening the user must ensure that there are no hazardous fluids or media are remaining inside the Heat Exchanger and care is taken for sufficient venting (supply of oxygen/breathable air).
After each opening of flanges the gaskets, stud bolts and nuts shall be checked and replaced in case of damages.
Before the Channel is closed the gasket surface the gasket and the threads of the bolts have to be checked for cleanliness and cleaned if necessary.

Tightening of bolts of Channel Flanges

The bolts shall be tightened by controlled torque with a calibrated torque wrench:

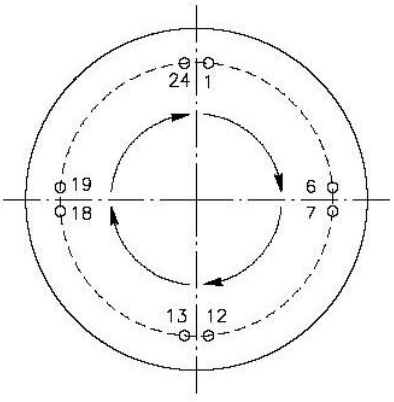
Torque per bolt	:	600 Nm / bolt
no. of bolts	:	68 1" UNC
sequence of applying the torque	:	see sketch on the right side. 1, 18, 35, 52 // 2, 19, 36, 53 // 3, 20, 37, 54 // 4, 21, 38, 55 // … 14, 31, 48, 65 // 15, 32, 49, 66 // 16, 33, 50, 67 // 17, 34, 51, 68
no of cycles	:	minimum 3
greasing of threads & bearing areas	:	MOLYKOTE 1000 or similar



Tightening of bolts of Interconnecting nozzles T2/T3

The bolts shall be tightened by controlled torque with a calibrated torque wrench:

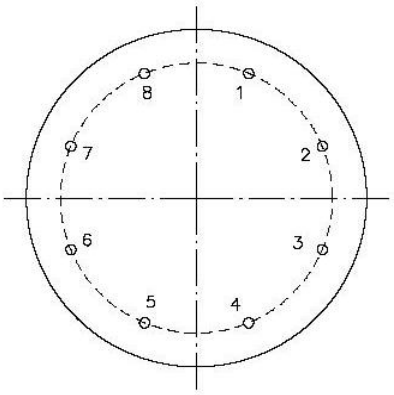
Torque per bolt	:	1200 Nm / bolt (350 kN)
no. of bolts	:	24 1 1/2" 8UN
sequence of applying the torque	:	see sketch on the right side. 1, 7, 13, 19 // 2, 8, 14, 20 // 3, 9, 15, 21 // 4, 10, 16, 22 // 5, 11, 17, 23 // 6, 12, 18, 24
no of cycles	:	minimum 3
greasing of threads & bearing areas	:	MOLYKOTE 1000 or similar



Tightening of bolts of Interconnecting nozzles S2/S3

The bolts shall be tightened by controlled torque with a calibrated torque wrench:

Torque per bolt	:	200 Nm / bolt
no. of bolts	:	8 3/4" UNC
sequence of applying the torque	:	see sketch on the right side. 1, 3, 5, 7 // 2, 4, 6, 8
no of cycles	:	minimum 3
greasing of threads & bearing areas	:	MOLYKOTE 1000 or similar



Inspection during operation

Because of changes of temperature and the seating behaviour of the gasket it may be necessary to retighten the bolts after the first installation.
If retightening is necessary it is permitted only if the Heat Exchanger is depressurised and cooled down to ambient temperature.

see:
- GA-drawing

Inspection

With the material chosen and the design data given there are no restrictions to shorten the interval for regular inspections below the limits required by law. If there are no limits by law, Hans Leffer GmbH & Co. KG recommends regular inspections in the intervals of:

Out-Side Inspection: 2 Years
In-Side Inspection: 5 Years
Pressure Test: 10 Years